

Q10. Concept Map – DBMS Architecture

Q10

DBMS ARCHITECTURE

DBMS Architecture refers to the **overall structure and organization** of a Database Management System. It shows how different components work together to **manage data** and **provide services** to users.

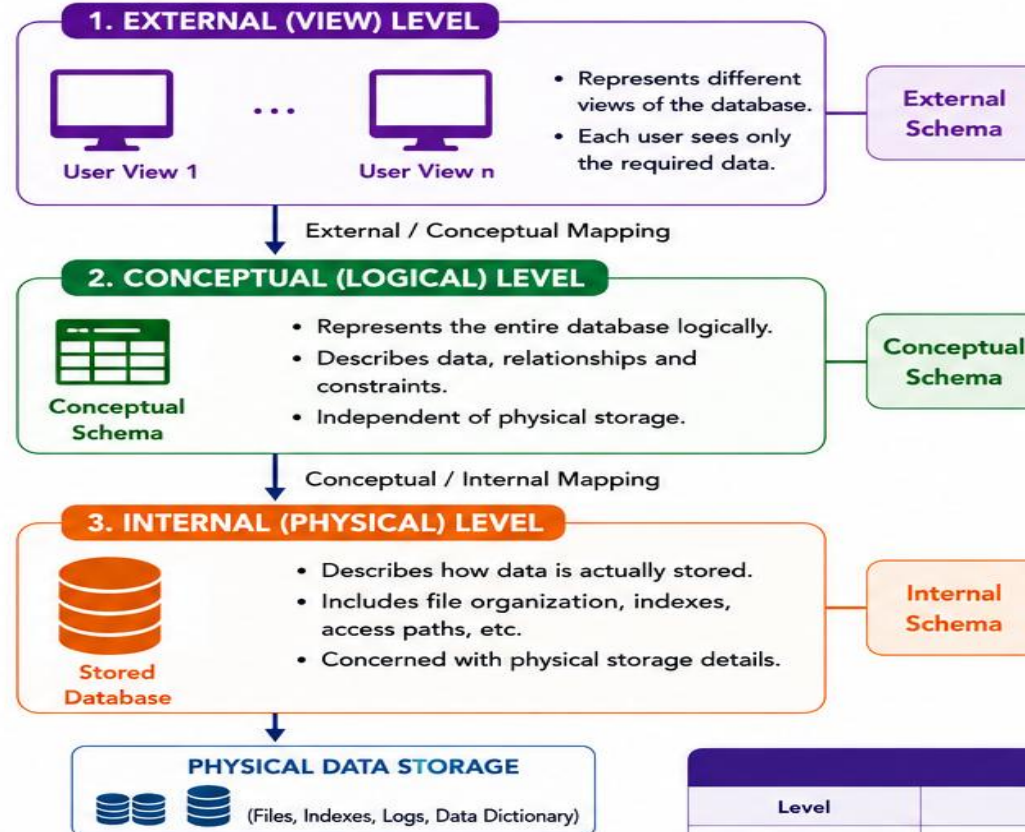
FEATURES OF DBMS ARCHITECTURE

- ✓ **Data Independence**
Changes in one level do not affect other levels.
- ✓ **Data Abstraction**
Hides unnecessary details from users.
- ✓ **Multiple User Views**
Different users can have different views of the same data.
- ✓ **Security**
Controls access to data and ensures data privacy.
- ✓ **Integrity & Consistency**
Maintains accuracy and consistency of data.

MAPPING IN DBMS ARCHITECTURE

- **External / Conceptual Mapping**
Maps user views to the conceptual schema.
- **Conceptual / Internal Mapping**
Maps the conceptual schema to the internal (physical) schema.

THREE LEVEL ARCHITECTURE OF DBMS



COMPONENTS OF DBMS ARCHITECTURE

	Users	People who interact with the DBMS.
	Application Programs	Programs that access the database through DDL/DML commands.
	Query Processor	Interprets and processes queries and requests.
	Storage Manager	Manages storage structure and access to data on disk.
	Data Dictionary	Stores metadata about data, structures, constraints, etc.

ADVANTAGES OF DBMS ARCHITECTURE

- ★ Provides a clear structure for database management.
- ★ Offers data independence and flexibility.
- ★ Supports multiple user views and applications.
- ★ Improves security and data integrity.
- ★ Easier maintenance and scalability.

LEVELS AT A GLANCE

Level	Purpose	Schema	Independence Provided
External Level	User views of data	External Schema	Physical Data Independence
Conceptual Level	Logical structure of entire database	Conceptual Schema	Logical Data Independence
Internal Level	Physical storage of data	Internal Schema	—

DBMS Architecture (Three Level Architecture) consists of External, Conceptual and Internal levels. It provides data independence, abstraction and efficient management of data. All components work together to provide a reliable, secure and efficient database system.